

Grassland Vs Desert Ecosystem: Productivity Comparison

Date: (08/04/2026)

Name of the presenter: Patel Tirth Navinbhai

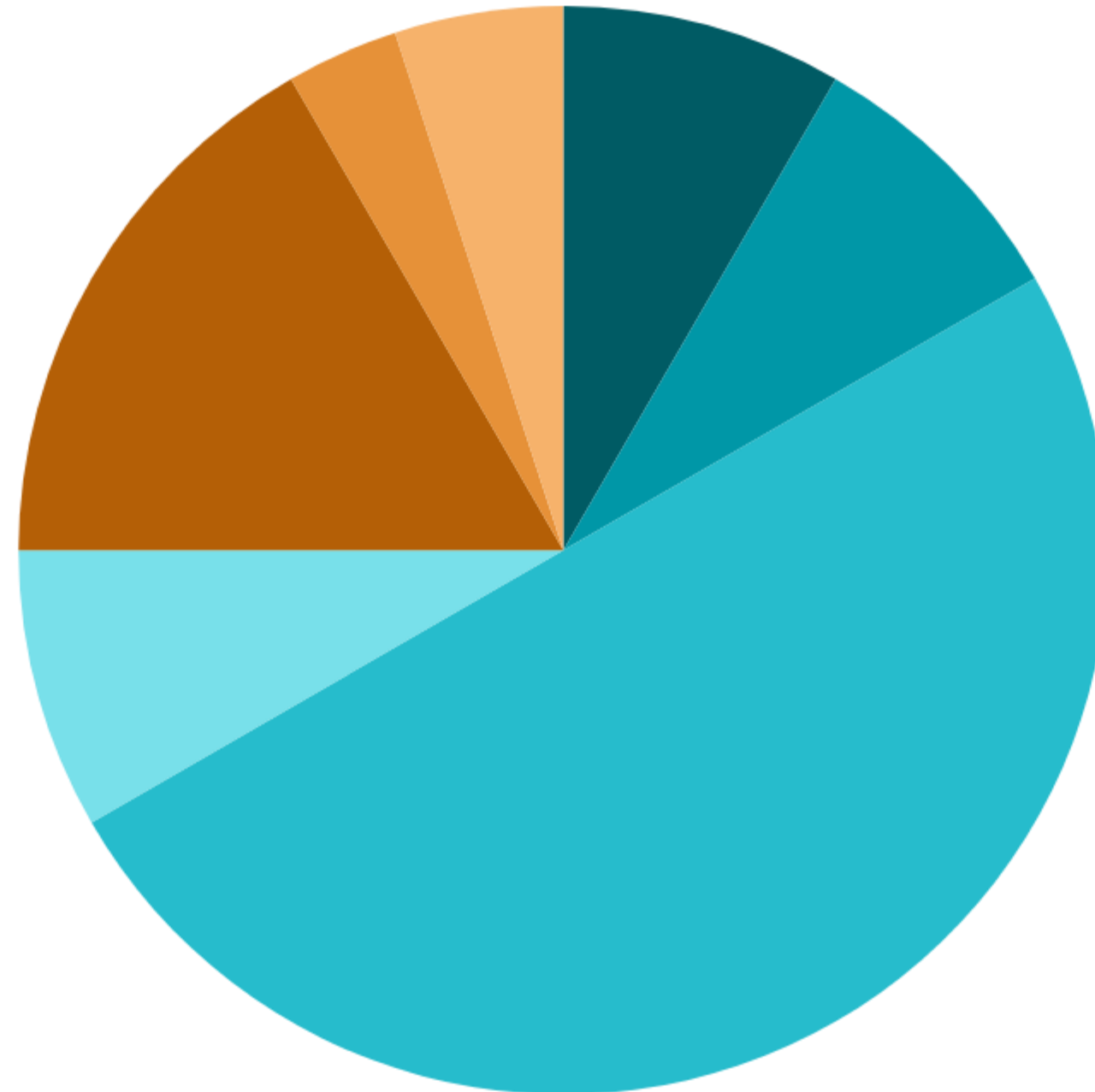
Department: BSc Data Science Sem-IV



Outline

- 1.What is an Ecosystem?
- 2.What is Productivity & How it is measured?
- 3.What is Grassland ecosystem?
- 4.What is Desert Ecosystem?
- 5.Comparison between productivity
- 6.Why Productivity differs?
- 7.Conclusion

Lecture Structure



	05 Minutes	Outline & Objectives
	05 Minutes	Recap of Previous Lecture
	30 Minutes	Delivery of Actual Content as per Lesson Plan
	05 Minutes	Industry Relevance
	10 Minutes	Interactive Activities
	02 Minutes	Importance of Discipline
	03 Minutes	Recording of Attendance

Reference: Office Order No. Provost/092023135, Date: 18th September 2023

What is an Ecosystem?

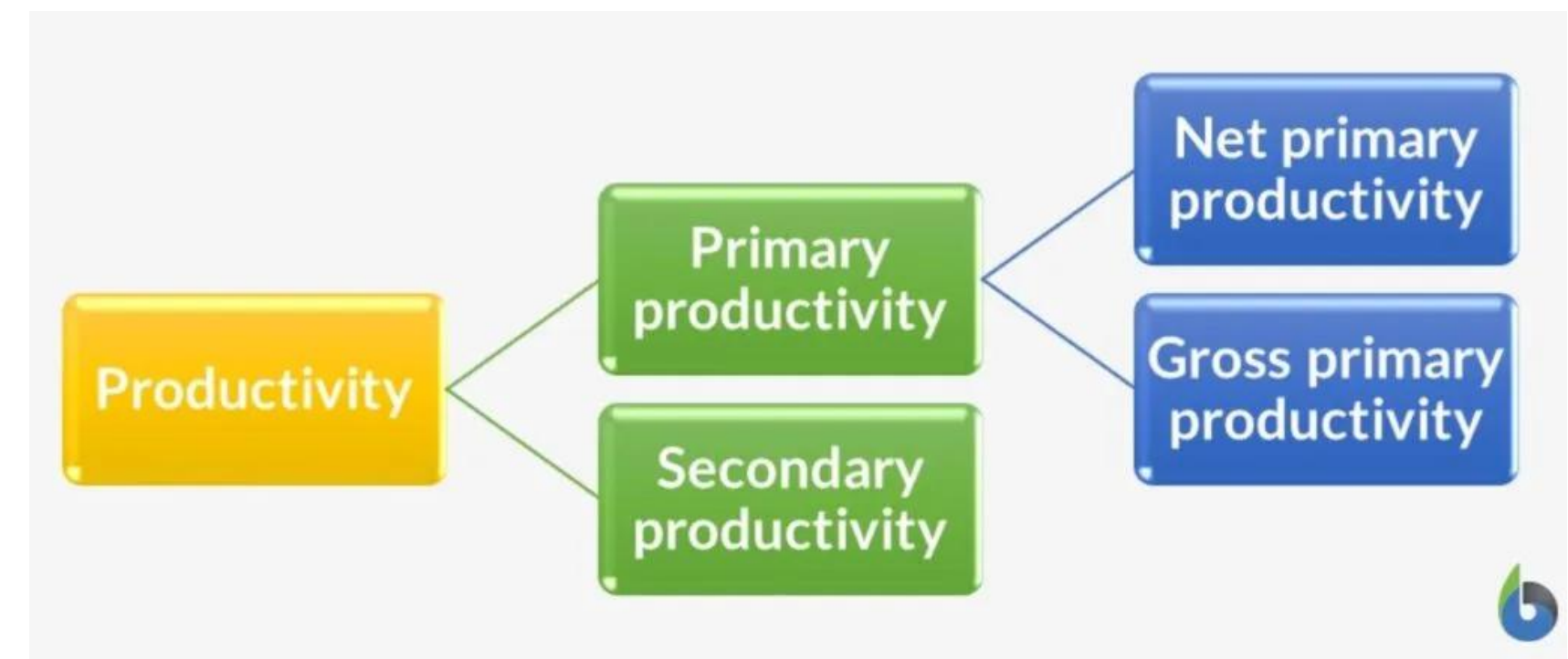
- Ecosystem is a dynamic community of living beings with their natural habitat & non living components of their environment.
- It is a network of interactions between biotic and abiotic components in a specific area such as forests, deserts, oceans etc etc.



Source of Image: AI Generated

What is Productivity?

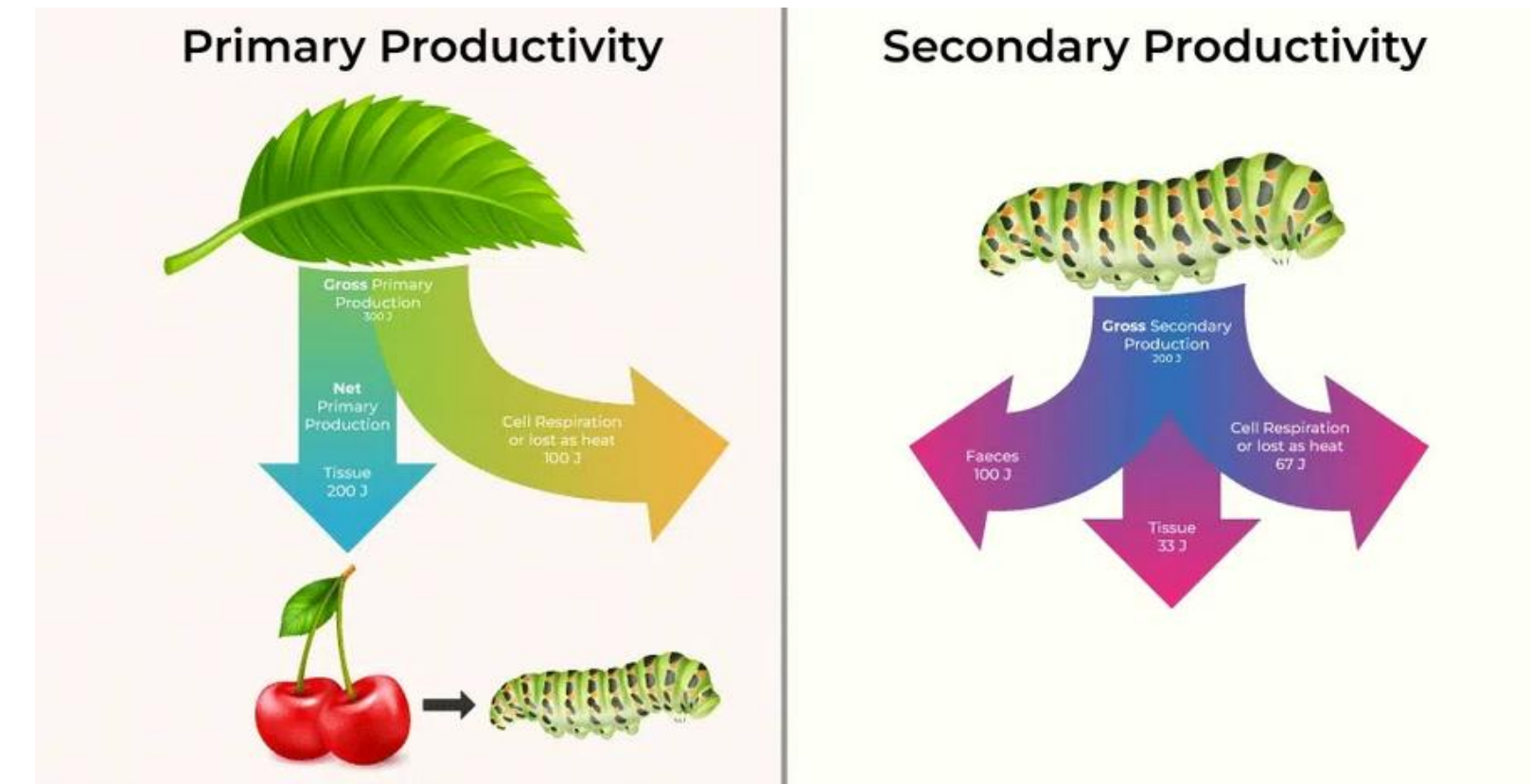
- Productivity refers to the rate at which energy or biomass is produced in an ecosystem. It determines how much energy is available to support life.
- There are two types of productivity, namely:
- **Primary Productivity**
- **Secondary Productivity**



Source of Image: <https://share.google/hYyXDb4liu0rz0vYb>

Types of Productivity

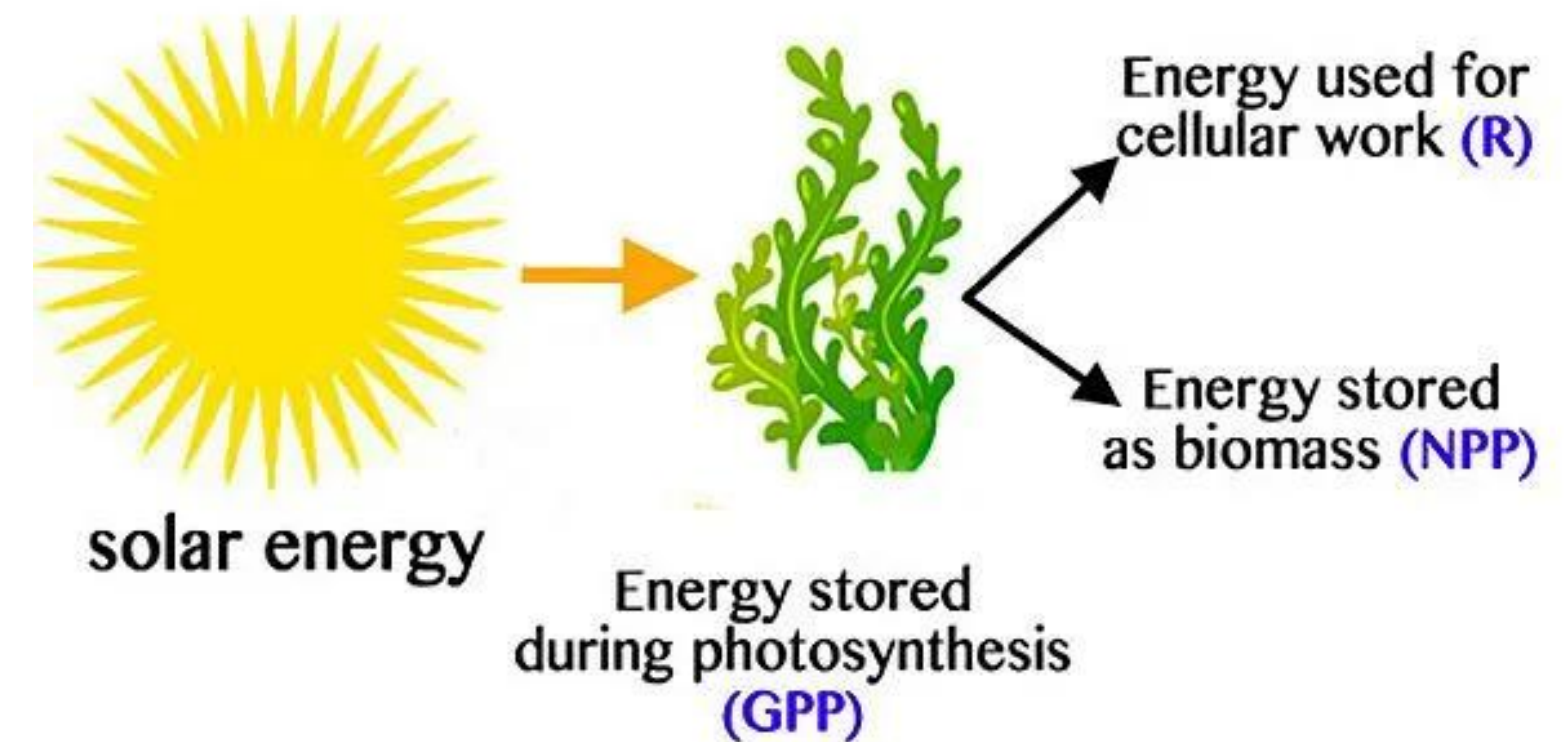
- **Primary Productivity:** It is produced by producers (plants, algae) through photosynthesis.
- This process of photosynthesis creates primary productivity which serves as the fundamental energy source for all food chains.
- **Secondary Productivity:** It is produced by consumers (animals, birds, insects)
- Animals consume plants and other organisms to produce their own energy. The process of energy transfer to higher trophic levels occurs through this mechanism.



Source of Image: <https://share.google/hYyXDb4liu0rz0vYb>

How it is measured?

- Ecosystem productivity is measured by analyzing the energy captured by plants and the energy remaining after respiration, which is available to consumers.
- It is measured with the help of formula: **$NPP = GPP - R$** ;
- **GPP (Gross Primary Productivity)**: Total energy produced by plants through photosynthesis
- **R (Respiration)**: The energy consumed by plants for their survival which includes growth, cellular respiration and maintenance
- **NPP (Net Primary Productivity)**: Energy left after **respiration**, which gets transfer to primary consumers(herbivores). It determines the productivity of ecosystem



$$NPP = GPP - R$$

net primary productivity gross primary productivity respiration

Source of Image: <https://cdn-ildfibj.nitrocdn.com/qaWeLntkRcyxNUcEoWbLpZLQkcglFsiJ/assets/images/optimized/rev-c337f6a/www.biologyonline.com/wp-content/uploads/2019/10/what-is-net-primary-productivity.jpg>

What is Grassland Ecosystem?

- Grassland ecosystem is a terrestrial biome dominated by grasses with very few trees
- It covers approximately 40% of Earth's land surface
- As for **vegetation**:
- Grasses are dominant
- Deep root systems help in fast regrowth
- **Animals:**
- Herbivores: Deer, Zebra, Bison
- Carnivores: Lion, Fox
- Adapted for grazing and running
- Grasslands support diverse life due to balanced environmental conditions and fertile soil.

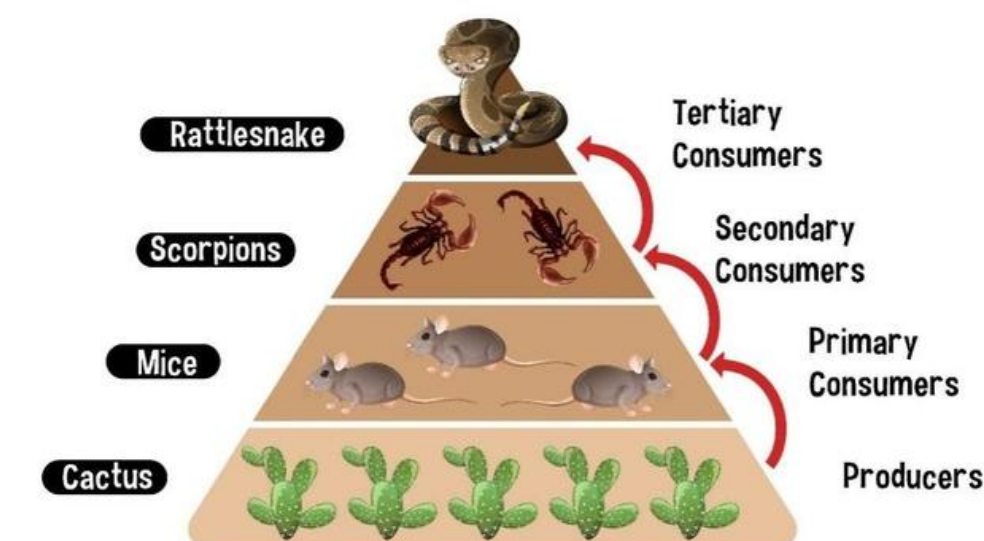
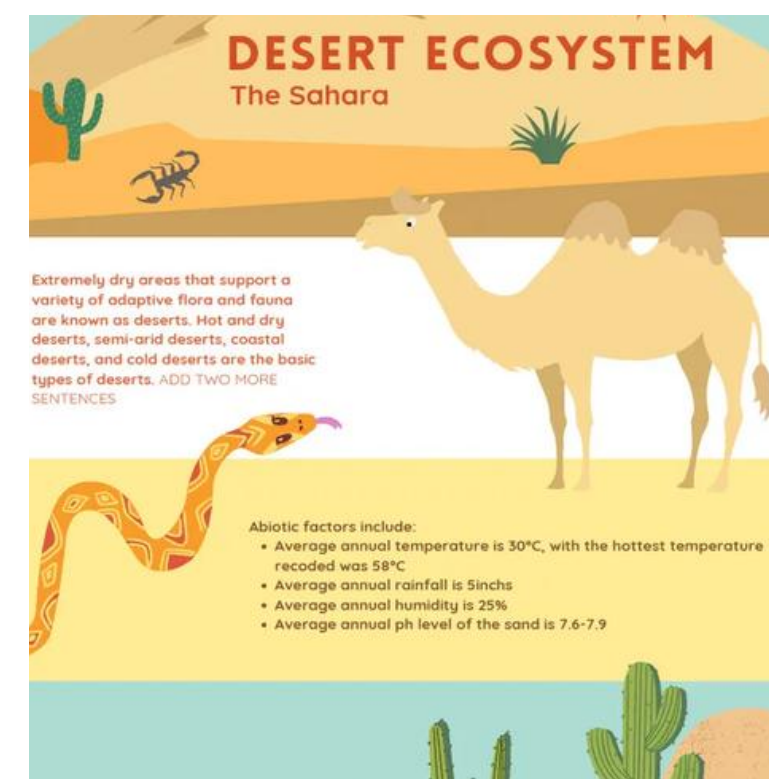


Source of Image: <https://share.google/hYyXDb4liu0rz0vYb>



What is Desert Ecosystem?

- Desert ecosystem is a terrestrial biome characterized by extremely low rainfall and sparse vegetation
- It covers approximately 33% of Earth's land surface
- It is mostly dominated by xerophytes (drought-resistant plants) such as cactus, acacia
- Desert plants use Crassulacean Acid Metabolism (CAM), it helps them to prevent water loss and increase survival in arid conditions
- Animals adapted to extreme heat and water scarcity.
- Examples: Camel, lizards, snakes, desert fox



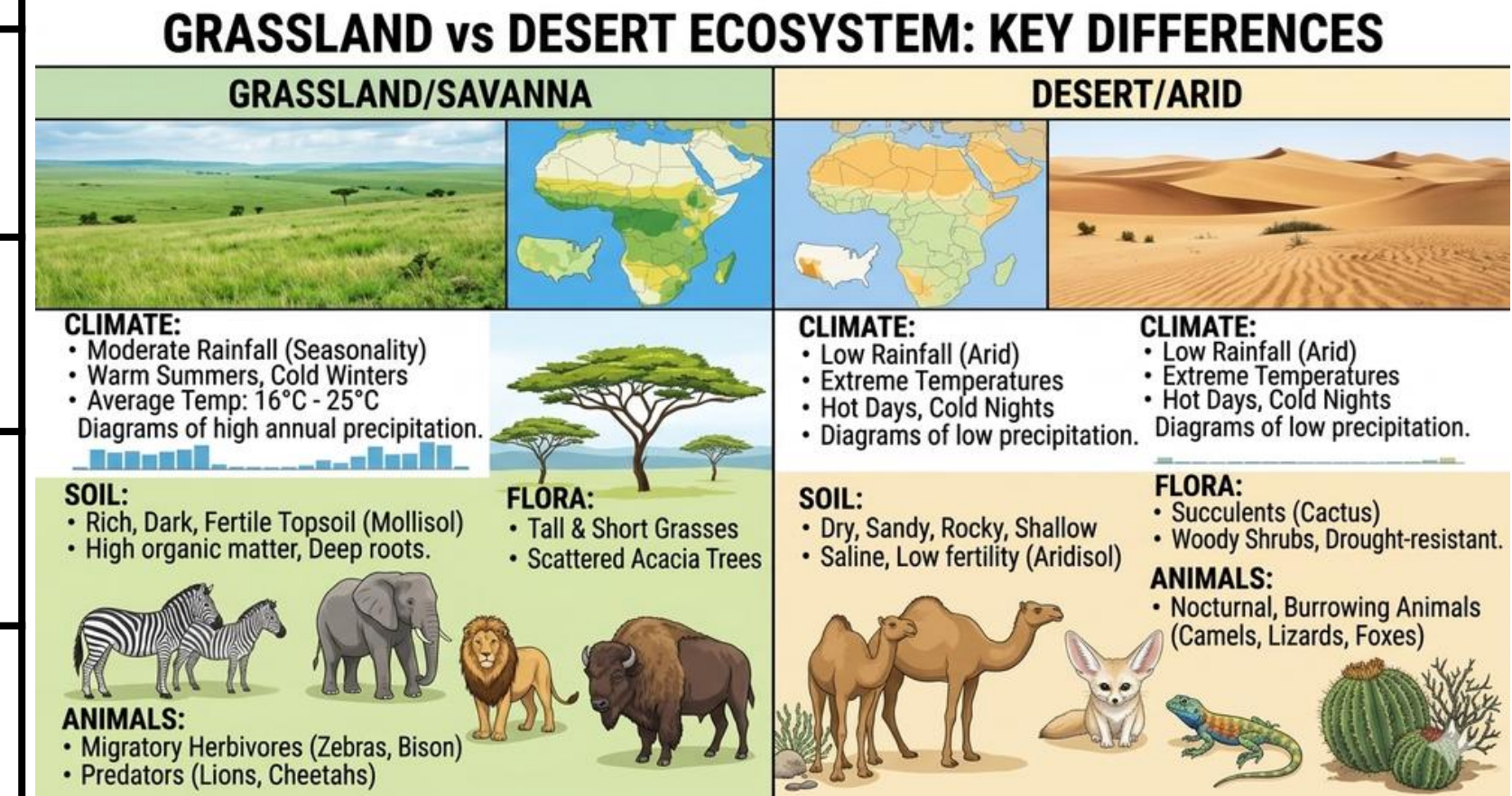
Source of Image:

<https://imgv21f.scribdassets.com/img/document/716975842/original/9a8=1>

<https://cmds-test.vedantu.com/prod/question-sets/597452ce-4c78-4aa7-3a8338f694765976638051711424511.png>

Comparison

Factor	Grassland	Desert
Rainfall	Moderate	Very Low
Vegetation	Dense	Sparse
Soil Fertility	High	Low
Productivity	Medium-High	Very Low
Biodiversity	High	Low



Reference: AI Generated

Water availability is the key factor that differentiates productivity between these ecosystems.”



Why Productivity differs?

Water Availability

- Grassland receives sufficient water
- Desert gets severe scarcity

Temperature

- Grassland has moderate conditions
- Desert has extreme conditions

Sunlight

- Both receive sunlight
- But desert lacks water to utilize it

Soil Quality

- Grassland has fertile soil
- Desert has poor nutrients

Productivity is controlled by limiting factors, especially water in desert ecosystems.



Conclusion

- Grasslands show higher productivity due to better environmental conditions
- Deserts have low productivity due to water scarcity and harsh climate
- Ecosystem productivity depends on environmental conditions, and even small changes in factors like water can significantly impact energy flow and life support.

Think-Pair-Share

Details of Interactive Activity



Image:
<https://i.pinimg.com/originals/9f/0e/5f/9f0e5fee28378215a04056924aca8336.jpg>

Reference: Please write / cite the references used (If any) for this slide (Font: Verdana, Size: 09, Color: Black)



GSFCU & Values

In every lecture, Faculty may talk on any of the following:

- **DCS & GSFCU Portal:** Remind students to use DCS, talk about it's features and instruct all to visit GSFCU website
- **Discipline, Values and Ethics.**
- **GSFC University – My University:** Emphasize the importance of taking ownership of the University.
- **Respect & Courtesy:** Remind students to be respectful to peers, teachers, and staff. Encourage polite language and behavior.
- **Punctuality:** Emphasize the importance of arriving on time to class and completing assignments by deadlines.
- **Classroom Rules:** Reinforce basic classroom rules, such as keeping mobile phones on silent and maintaining a clean environment.
- **Safety:** Remind students about safety protocols, whether it's physical safety in the classroom or digital safety when using online resources.
- **Positive Attitude:** Encourage a positive mindset, cooperation and supporting one another.

Any other thoughts related to Discipline, Values, Ethics etc. can be mentioned during these 2 minutes. You May use appropriate image / video with proper reference. One example is given in next slide.

Reference: Please write / cite the references used (If any) for this slide (Font: Verdana, Size: 09, Color: Black)

 **Thank You..**